

Description of the Cleaned NHANES Dataset

Josephine Abena Agyakwa Asare

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1 Introduction

The National Health and Nutrition Examination Survey (NHANES) is a program that collects health and nutrition information from the U.S population (Centers for Disease Control and Prevention, n.d.). This report provides a descriptive summary of the cleaned NHANES dataset used in this project.

The tables below describe the sample across NHANES waves, including sample size, sex distribution, and ethnicity categories. Summary statistics are also presented for average systolic blood pressure (SBP) by wave.

2 Data Preparation

```
library(tidyverse)
library(rio)
library(here)
library(knitr)
library(kableExtra)

nhanes_data <- import(here("data", "cleaned_NHANES.csv"))
```

3 Sample Size by Wave

The table below shows the overall number of participants included in each NHANES wave. This provides an overview of the size of the sample represented in each survey wave.

```
sample_size_by_wave <- nhanes_data %>%
  group_by(wave_id) %>%
  summarise(
    sample_size = n(),
    .groups = "drop"
  )

kable(
  sample_size_by_wave,
  format = "latex",
  booktabs = TRUE,
  caption = "Overall Sample Size by NHANES Wave",
  col.names = c("NHANES Wave", "Sample Size")
) %>%
  kable_styling(
    latex_options = "hold_position"
  )
```

Table 1: Overall Sample Size by NHANES Wave

NHANES Wave	Sample Size
2015	10175
2016	9971
2017	9254

4 Male Participants by Wave

The table below presents the number and percentage of male participants in each NHANES wave. Percentages are calculated within each wave so that the sex distribution can be compared across waves.

```
male_by_wave <- nhanes_data %>%
  group_by(wave_id) %>%
  summarise(
    sample_size = n(),
```

```

    male_n = sum(gender == "Male", na.rm = TRUE),
    male_percent = 100 * male_n / sample_size,
    .groups = "drop"
  ) %>%
  select(wave_id, male_n, male_percent)

kable(
  male_by_wave,
  format = "latex",
  booktabs = TRUE,
  digits = 1,
  caption = "Number and Percentage of Male Participants by NHANES Wave",
  col.names = c("NHANES Wave", "Male n", "Male (%)")
) %>%
  kable_styling(
    latex_options = "hold_position"
  ) %>%
  row_spec(
    0,
    bold = TRUE,
    background = "gray!20"
  )

```

Table 2: Number and Percentage of Male Participants by NHANES Wave

NHANES Wave	Male n	Male (%)
2015	5003	49.2
2016	4892	49.1
2017	4557	49.2

5 Ethnicity Distribution by Wave

The table below presents the number and percentage of participants in each ethnicity category within each NHANES wave. Percentages are calculated within each wave to allow comparison of the distribution across survey waves.

```

ethnicity_by_wave <- nhanes_data %>%
  group_by(wave_id, ethnicity_2) %>%
  summarise(
    n = n(),
    .groups = "drop"
  ) %>%
  group_by(wave_id) %>%
  mutate(
    percent = 100 * n / sum(n)
  ) %>%
  ungroup()

kable(

```

```

ethnicity_by_wave,
format = "latex",
booktabs = TRUE,
digits = 1,
caption = "Ethnicity Distribution by NHANES Wave",
col.names = c("NHANES Wave", "Ethnicity", "n", "Percent")
) %>%
kable_styling(
  latex_options = "hold_position"
) %>%
row_spec(
  0,
  bold = TRUE,
  background = "gray!20"
)

```

Table 3: Ethnicity Distribution by NHANES Wave

NHANES Wave	Ethnicity	n	Percent
2015	Asian	1074	10.6
2015	Black	2267	22.3
2015	Mexican	1730	17.0
2015	Other	470	4.6
2015	Other Hispanic	960	9.4
2015	White	3674	36.1
2016	Asian	1042	10.5
2016	Black	2129	21.4
2016	Mexican	1921	19.3
2016	Other	505	5.1
2016	Other Hispanic	1308	13.1
2016	White	3066	30.7
2017	Asian	1168	12.6
2017	Black	2115	22.9
2017	Mexican	1367	14.8
2017	Other	634	6.9
2017	Other Hispanic	820	8.9
2017	White	3150	34.0

6 Average Systolic Blood Pressure by Wave

Average systolic blood pressure was calculated from the four available systolic blood pressure measurements for each participant. The table summarizes the distribution of average systolic blood pressure within each NHANES wave.

```

nhanes_data <- nhanes_data %>%
  mutate(
    average_sbp = rowMeans(
      select(

```

```

    .,
    systolic_bp_1,
    systolic_bp_2,
    systolic_bp_3,
    systolic_bp_4
  ),
  na.rm = TRUE
),
average_sbp = ifelse(
  is.nan(average_sbp),
  NA,
  average_sbp
)
)

```

7 Average Systolic Blood Pressure by Wave

Average systolic blood pressure was calculated from the four available systolic blood pressure measurements for each participant. The table summarizes the distribution of average systolic blood pressure within each NHANES wave.

```

sbp_by_wave <- nhanes_data %>%
  group_by(wave_id) %>%
  summarise(
    minimum = min(average_sbp, na.rm = TRUE),
    maximum = max(average_sbp, na.rm = TRUE),
    median = median(average_sbp, na.rm = TRUE),
    mean = mean(average_sbp, na.rm = TRUE),
    standard_deviation = sd(average_sbp, na.rm = TRUE),
    .groups = "drop"
  )

kable(
  sbp_by_wave,
  format = "latex",
  booktabs = TRUE,
  digits = 1,
  caption = "Summary Statistics of Average Systolic Blood Pressure by NHANES Wave",
  col.names = c(
    "NHANES Wave",
    "Minimum",
    "Maximum",
    "Median",
    "Mean",
    "Standard Deviation"
  )
) %>%
kable_styling(
  latex_options = "hold_position"
) %>%
row_spec(
  0,

```

```
bold = TRUE,
background = "gray!20"
)
```

Table 4: Summary Statistics of Average Systolic Blood Pressure by NHANES Wave

NHANES Wave	Minimum	Maximum	Median	Mean	Standard Deviation
2015	64.7	228.7	115.3	118.3	18.1
2016	74.0	231.3	117.3	120.4	18.4
2017	72.7	234.0	118.0	121.7	20.3

8 Summary of Findings

The NHANES sample is distributed across multiple survey waves, with the overall sample size varying between waves. The sample size table provides an overview of the number of participants represented in each wave.

The proportion of male participants also varies across waves. The male participant table presents both the number and percentage of males within each wave, allowing the sex distribution to be compared across survey periods.

The ethnicity table shows the distribution of participants across the cleaned ethnicity categories within each wave. Presenting both counts and percentages makes it possible to compare the composition of the sample across waves while accounting for differences in sample size.

Average systolic blood pressure also varies across the NHANES waves. The summary statistics table presents the minimum, maximum, median, mean, and standard deviation of average systolic blood pressure for each wave. These measures describe both the central tendency and variability of systolic blood pressure in the sample.

References

Centers for Disease Control and Prevention. n.d. "National Health and Nutrition Examination Survey Data." Hyattsville, MD: U.S. Department of Health; Human Services; National Center for Health Statistics. <https://wwwn.cdc.gov/nchs/nhanes/>.